Brooke Yang

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EDUCATION

University of Waterloo — Waterloo, Ontario
Bachelor of Mathematics, Combinatorics & Optimization (Pure Math Joint) — *September 2023 – April 2028*

EXPERIENCE

Cerebras Systems — Toronto, Ontario
Inference Service Automation Engineer Intern — *May 2026 – August 2026*

- Built a Loki log triaging tool that **reduced release qualification time by ~83%** by automating log deduplication, error clustering, and RCA verdict caching
- Developed a **keyless ECR pipeline** (GitHub Actions OIDC) to validate control-plane images in EKS clusters, serving as the automated source of stable image tags for **all platform deployments**

Huawei Canada — Waterloo, Ontario
Assistant Engineer Intern — *January 2025 – August 2025*

- Co-authored an *Agentic Systems Reference Architecture* adopted by **12+ teams** across Huawei Research Labs, featuring low-level specifications for MCP tooling, VM-sandboxed execution, and MARL
- Designed **multi-agent architectures** for dynamic 5G networks, orchestrating automated **CLI scripting** to reduce operational overhead on Policy Control Functions by **up to 35%**
- Conducted **40+ IR experiments** on asymmetric semantic search, leveraging HyDE and Cross-Encoder reranking to achieve an **18% MRR increase** for long-horizon multi-agent tasks
- Developed a logistic regression classifier to detect LLM hallucinations with **81.1% accuracy**, mapping token-distribution Shannon entropy to a sigmoid decision boundary

Remitbee — Mississauga, Ontario
AI Research Intern — *May 2024 – August 2024*

- Built a speech-to-text (STT) task automation API in TypeScript deployed on AWS Lambda, using audio segmentation to reduce **downstream service latency by 5x**
- Developed **NLP pipelines** for PII detection and normalization using Atlas vector search and fuzzy matching, achieving a **7% word error rate** (WER)
- Implemented CI/CD tests using BERTScore metrics to detect STT/NLP model regressions

PROJECTS

Fast Multipole Method (FMM) Solver | C++ — [Link]

- Implemented a 3D N-body solver from the *Greengard-Rokhlin paper*, using multipole translations (M2M, M2L, L2L) to **reduce potential field evaluation from $O(N^2)$ to $O(N)$**
- Achieved a **7.1x speedup** over direct summation at 128K particles, validated to ~1e-4 relative error

Multilateral Netting Engine | Python — [Link]

- Modelled settlement netting as a directed obligation graph, implementing greedy cycle-cancellation and exact Linear Programs (LPs in CVXPY) to **reduce gross notional value by 85%**
- Quantified residual exposure under per-counterparty credit limits using a penalty-based LP relaxation

Noisy Keynesian Guessing Game | Python — [Link]

- Trained **PPO agents** with feedforward and recurrent policies in a multi-agent environment under Gaussian noise models to study convergence behavior
- Demonstrated **non-monotonic noise effects** on Nash equilibrium convergence, with an **8.0 SD** outperforming lower and higher noise levels

SKILLS

Languages: Python, C++, C, Bash, SQL, TypeScript, JavaScript, Java, R, Haskell
Tools/frameworks: Kubernetes, Docker, AWS (EKS, ECR, S3), PyTorch, ArgoCD, Jenkins, Bazel, LangChain, Boto3, Grafana/Loki, NumPy, Pandas, MongoDB, SQLite, Matplotlib, React, Git

HONORS & AWARDS

- MEF Recipient, Canadian Undergraduate Mathematics Conference (CUMC) — 2026
- Top 5 Finalist, IMC Trading US Chess Academy Qualifier — 2026
- AFM1st Peel Region Summer 3 Chess Champion — 2025
- CCC Senior, Certificate of Distinction and School Champion — 2023
- Euclid Contest, Certificate of Distinction and School Champion — 2022, 2023